Tools 4.2: Interests and Rational Choice — Why Do Political Actors Choose What They Choose? (AD)

Political Science — Master Notes · by Jakob V. Stangel

Required reading: Bueno de Mesquita, *Political Economy for Public Policy*, **ch. 2.1** · *Optional:* **Kenneth Shepsle**, *Analyzing Politics: Rationality, Behavior, and Institutions*, **ch. 2**. *(Lecturer: Anna Denisenko — deck "4.2 Interests and Rational Choice.")*

Theme of the lecture. Lecture 4.1 asked the **normative** question (which outcomes are *good*); this one asks the **positive** question (**why** actors choose what they choose, and how those choices **add up to outcomes**). The tool is **rational choice**: *political and business outcomes emerge from actors pursuing their goals under constraints.* Build any analysis from **three blocks** — **actors, preferences, constraints** — and you can predict behaviour, see why "good-on-paper" policies fail, and know **which lever** (goals, incentives, or available actions) actually changes an outcome. When one actor's best move depends on others', we add **game theory** and the **Nash equilibrium** — and meet the **prisoner's dilemma**, where individually rational choices produce a collectively bad result.

> ★ **The heart of the lecture — outcomes = actors pursuing goals under constraints.**
> • **Three building blocks: actors** (who chooses?), **preferences** (a ranking of *outcomes*, written x ≻ y), and **constraints** (which options are actually feasible — resources, institutions, information).
> • **"Rational" = complete + transitive preferences** — *not* selfish, not only about money, not all-knowing, not "smart." Just a consistent ranking, pursued as best one can given beliefs and constraints.
> • **To change behaviour, change incentives** (or the available actions) — *not* people's goals. An **incentive** is anything that shifts the costs/benefits of an outcome, and **institutions shape incentives.**
> • **Strategic situations → game theory.** When your best choice depends on what others do, find the **Nash equilibrium**: a strategy profile where everyone is best-responding, so no one wants to deviate. The housing reform and the trade war are both **prisoner's dilemmas** — individually rational, collectively bad.

Red thread: *(1) Who are the actors, what do they want, and what constrains them? (2) Which lever changes the outcome? (3) When choices are interdependent, what's the stable (Nash) result — and is it good?*

Key concepts to apply: *rational choice (outcomes from goals + constraints) · actors · preferences (ranking over **outcomes**, x ≻ y) · constraints (resources · institutions · information) · rationality = **complete + transitive** · what rationality is **not** · the paradox of voting · expressive/duty voting · incentives · the three levers (goals · incentives · actions) · Mayhew (re-election-seeking politicians) · Niskanen (budget-maximising bureaucrats) · **expected value** & when cheating pays · game theory: players · strategies · timing · payoffs · payoff matrix · best response · **Nash equilibrium** · the **prisoner's dilemma** (individually rational → collectively bad; not Pareto efficient) · (critiques) bounded rationality (Simon) · loss aversion / status-quo bias (Kahneman & Tversky) · preferences taken as given · "model as map."*

From "should" to "why"

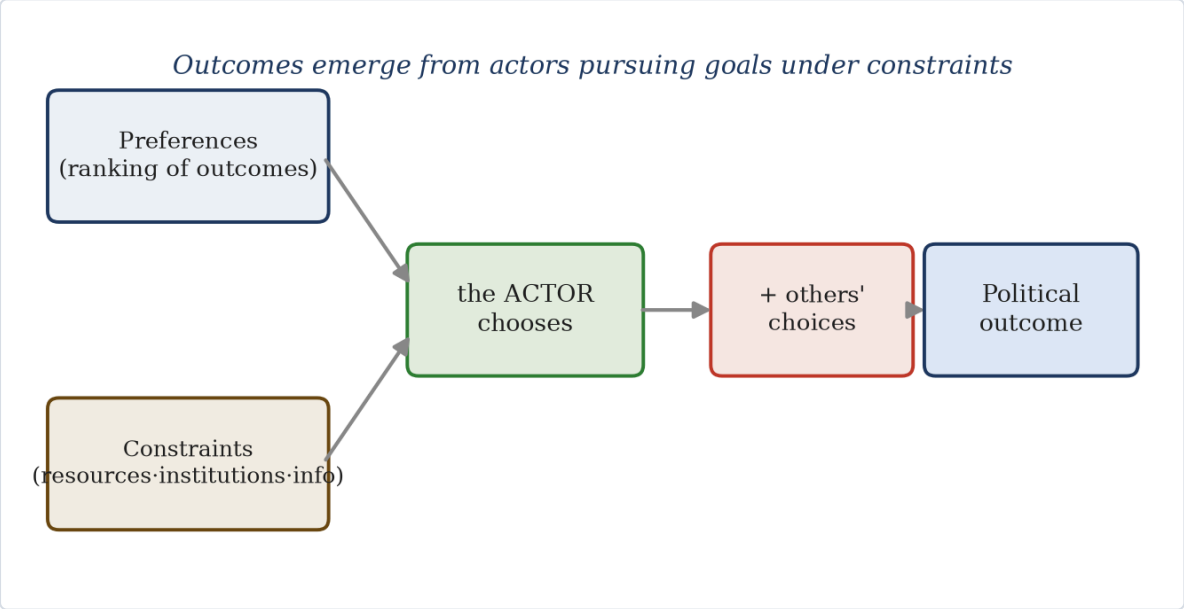
Core: normative frameworks tell us which outcomes are **good**; they do **not** tell us which outcomes will **actually happen.** Rational choice fills that gap.

The lecture's motivating **puzzles** — all answerable with one tool:

- Your single vote **almost never** decides an election — so **why do millions vote**?
- Why do **firms break rules** even when fines are large?
- Why might **abolishing a height limit fail** to fix a housing shortage?
- Why do countries **raise tariffs** when they know free trade makes everyone better off?

★ The three building blocks

Core: *political/business outcomes emerge from actors pursuing their goals under constraints.* Build any analysis from three pieces.



How to read it. An **actor** ranks outcomes (**preferences**) and faces **constraints** (what's feasible); from those they make a **choice**; that choice **plus others' choices** produces the **political outcome.** Change any block and you change the prediction.

1. **Actors** — who is making the choice?
2. **Preferences** — what outcomes do they want?
3. **Constraints** — what options are actually available?

Actors & preferences

Core: we name the actor, assume a **simple goal**, and write their **ranking of outcomes.**

Typical actor	Assumed preferences
Voters	good policy, low taxes, a party they like
Politicians	winning (re-)election, power
Bureaucrats	budget, autonomy, career
Firms	profit
Interest groups	policy wins for their members

- Sometimes we treat a **group** (a city, firm, country) as a **single actor** — a simplification we should **flag.**
- **Preferences = a ranking of possible outcomes.** Write **x ≻ y** = "the actor prefers x to y." E.g. a voter: *carbon tax ≻ green subsidies ≻ do nothing.*
- **Preferences are about outcomes, not actions.** (You don't prefer "voting"; you prefer the *outcome* voting helps bring about.) What shapes them: **goals, identity, values, incentives, information.**

★ What "rational" means

Core: a precise, modest definition — **not** the everyday sense of "clever" or "selfish."

Preferences are **rational** if they are:

- **Complete** — the actor can compare *any* two outcomes (x ≻ y, or y ≻ x, or indifferent).
- **Transitive** — the ranking is consistent: if x ≻ y and y ≻ z, then **x ≻ z.**

What rationality does *not* mean: not selfish (you can prefer helping others — altruism, fairness, duty); **not only about money** (can be identity or ideology); **not all-knowing** (actors can hold *wrong* beliefs); **not "smart" or "good"** (it's about **consistency**, not wisdom).

> RATIONAL CHOICE
>
> acting to achieve one's **most-preferred outcome**, given one's **constraints** and **beliefs**

Reads: given a **complete + transitive** ranking, the actor picks the **feasible action** they believe will lead to the **best** outcome.

Why vote? (the paradox of voting)

Core: the first puzzle — if the *only* goal is to change the result, voting looks **irrational**, yet turnout is high. Resolving it shows preferences are about more than the result.

- The chance a single vote decides a US presidential election is **~1 in 60 million** (Gelman, Silver & Edlin), and voting costs time/effort — so purely *instrumentally* it doesn't pay.
- Yet **~2/3 of eligible Americans voted in 2024**, and **~84% of Danes** in the 2022 Folketing election.

- **Why it can still be rational:** people value **more than** changing the result — the **act itself** (expressing identity/values, doing one's **duty**, participating). Rational choice accommodates this: these are real entries in the preference ranking. *(Preferences over outcomes ≠ "only the election result.")*

Constraints

Core: actors choose the **best feasible** option — so the **constraint set** often decides the outcome.

Types of constraint:

- **Resources** — money, time, seats in parliament.
- **Institutions** — laws, constitutions, electoral rules, EU treaties.
- **Information** — what you know about options and consequences.

→ **Same preferences + different constraints → different choices.** (Some actions are simply off the table — e.g. *voting twice is illegal.*) This is why **institutions matter**: they set what's feasible.

Incentives & the three levers

Core: we can't choose *for* people — we can only change **what shapes their choices.** There are exactly **three levers.**

Lever	How it works	Feasibility
1. Change goals	persuade people to *want* different things	slow and hard
2. Change incentives	shift the **costs/benefits** of outcomes	the usual policy tool
3. Change available actions	alter the **constraint set** (ban/allow options)	institutional design

- An **incentive** = anything that changes the **costs or benefits** of an outcome. **Institutions shape incentives:** electoral rules shape what politicians promise; budget rules shape what bureaucrats ask for; **fines and inspections** shape whether firms comply.
- **Useful simplifying assumptions about actors:** politicians as "**single-minded seekers of re-election**" (Mayhew 1974) — explains spending aimed at swing voters and blame-avoidance before elections; bureaucrats as **budget-maximisers** (Niskanen 1971) — explains "use it or lose it" year-end spending. *These are useful if they **predict**, not because they're literally true of everyone.*

★ Expected value: when does cheating pay?

Core: to predict a choice under risk, compare the **expected** costs and benefits. This is how you design a rule that actually works.

The firm problem. A firm can secretly break an environmental rule: extra profit from breaking it = **1 million kr**; fine if caught = **5 million kr**; chance of being caught = **1 in 10.**

EXPECTED COST OF CHEATING

$$\mathrm{E}[\text{fine}] = \text{probability} \times \text{fine} = 0.1 \times 5{,}000{,}000 = \mathbf{500{,}000\ kr}$$

Reads: on average, cheating costs **500,000 kr** — but it *gains* **1,000,000 kr.** Since the gain (1m) **exceeds** the expected fine (0.5m), a profit-maximiser **cheats.**

How the government flips the decision — raise the expected cost above the gain, via either lever:

- **bigger fine:** need fine × 0.1 > 1m → **fine > 10m**; or
- **more inspections:** need prob × 5m > 1m → **prob > 0.2** (1 in 5).

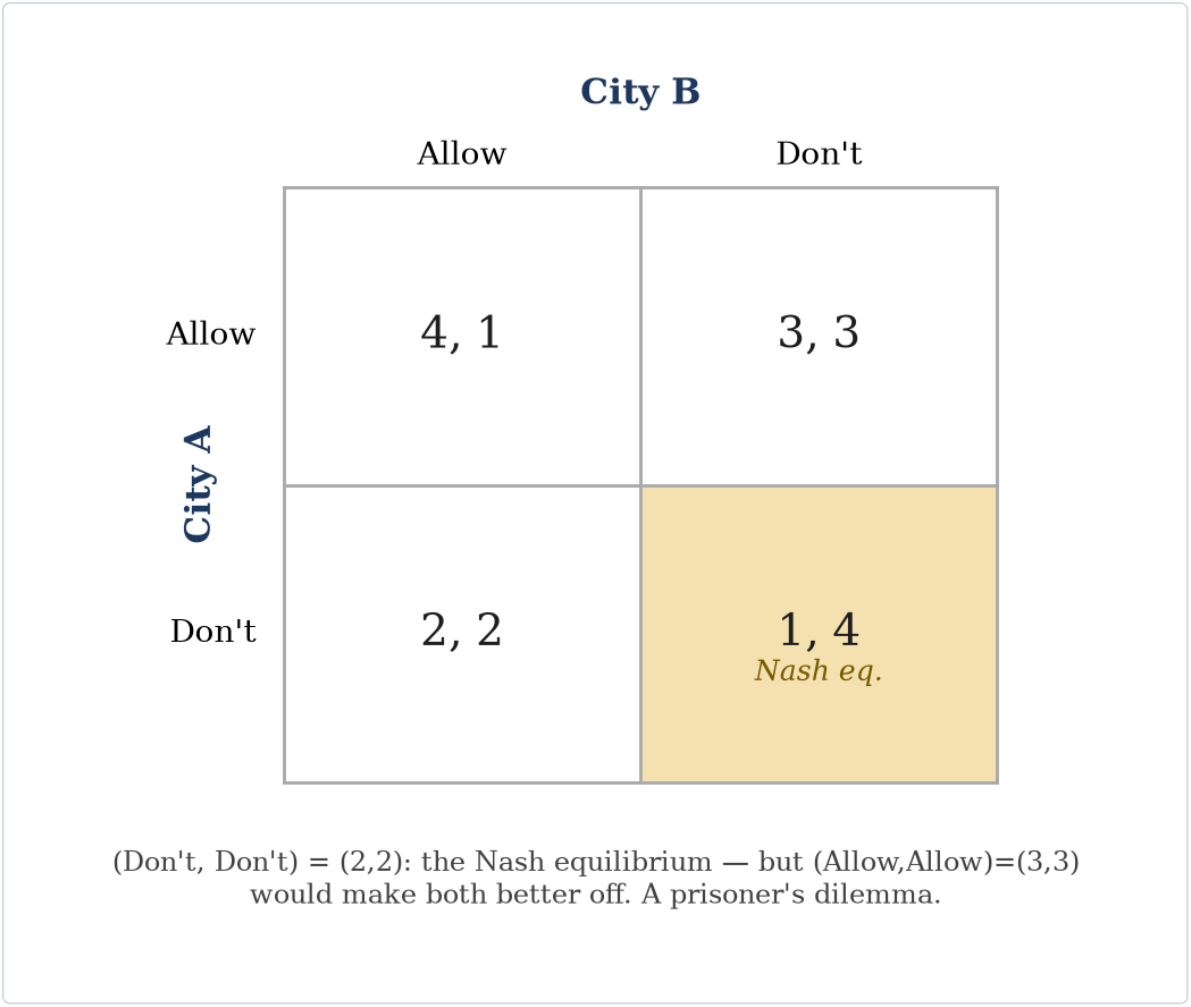
→ *The lesson:* "clean air is good" (normative) isn't enough — you need a **rule big enough that the actor chooses to comply** (rational choice). **Good intentions don't change behaviour; incentives do.**

★ Game theory & the Nash equilibrium

Core: when **your best choice depends on what others do**, you're in a **strategic situation** — use **game theory.** Every game answers **four questions**, and the prediction is the **Nash equilibrium.**

The four questions of a game: (1) Players — who's involved? **(2) Strategies** — what can each do? **(3) Timing** — who moves when? **(4) Payoffs** — what does each want?

The housing game. A national height ban causes a housing shortage; the reform lifts it but lets **each city** choose. Will it work? Model **two cities** (A, B), each can **Allow** tall buildings or **Don't**; they move **simultaneously**; each would rather **not** have towers itself but wants homelessness solved (and cares more about avoiding towers). Ranking → payoffs (higher = better; only the order matters):



(Don't, Don't) = (2,2): the Nash equilibrium — but (Allow,Allow)=(3,3) would make both better off. A prisoner's dilemma.

How to read it. A picks a **row**, B a **column**; each cell is (A's payoff, B's payoff).

- **Best response** = your best strategy given what the other does. *If B allows,* A prefers **Don't** (4 > 3); *if B doesn't,* A still prefers **Don't** (2 > 1). So "**Don't" is best for A no matter what** (same for B).
- **Nash equilibrium** = a **strategy profile where every player is best-responding**, so **no one wants to change alone.** Here that's **(Don't, Don't) = (2, 2)** — the highlighted cell.
- **The recipe:** for each player, **underline the best-response payoff** in every column/row; the cell where **all** payoffs are underlined is the **Nash equilibrium.** (A game can have one, several, or none of this simple kind.)
- **Why it's a good prediction:** it's the "**no regrets**" outcome — after seeing what others did, no one wishes they'd chosen differently. Non-equilibria are **unstable** (someone would switch). *(Named after **John Nash**, Nobel 1994 — "A Beautiful Mind.")*

→ **So will the reform fix the shortage?** No — both cities rationally choose **Don't**, and the towers never get built.

★ The prisoner's dilemma

Core: the housing game and the trade war are both a **prisoner's dilemma** — the signature result where **individually rational choices lead to a collectively bad outcome.**

- **The structure:** each player has a strong individual reason to pick the "selfish" option (**Don't / Tariff**); when **everyone** does, **everyone ends up worse off.**
- **The trade war** (same matrix): two countries choose **Free trade** or **Tariff**; the Nash equilibrium is **(Tariff, Tariff) = (2,2)**, though **(Free trade, Free trade) = (3,3)** is better for both. *Real life, 2025:* the US and China escalated tariffs to **145% / 125%**, then agreed a 90-day truce cutting them to **30% / 10%** — an attempt to climb back to the top-left cell.
- **Link to Lecture 4.1:** the equilibrium **(2,2) is *not* Pareto efficient** — moving to **(3,3)** would make *everyone* better off, with no loser. So a **Pareto** welfarist condemns the equilibrium; a **utilitarian** too (total 4 < 6); and the puzzle "**how do groups escape this trap?**" is **next lecture** (collective action).

Criticisms of rational choice (and why use it anyway)

Core: rational choice is a **model** — deliberately simplified. Know its limits, and why it's still worth using.

Criticisms:

- **Bounded rationality (Herbert Simon):** people have **limited time & information**, so they use **shortcuts** and settle for "good enough" (*satisficing*), not optimising.
- **Psychology — loss aversion (Kahneman & Tversky):** people weigh **losses about twice as heavily as equal gains**, so they **defend the status quo.** *Politics:* cutting an existing benefit causes **more anger** than never giving it — a **status-quo bias** rational choice misses.
- **Where do preferences come from?** Rational choice **takes them as given** (the *Ideas & Identity* lecture opens that box).
- **Too simple?** Real situations have **many** players and options.

Why use it anyway: it makes **assumptions explicit** (who wants what, what constrains them); produces **testable predictions** (more inspections → more compliance); helps **design policy** (the housing reform looked good on paper — a simple game showed why it fails); and **explains surprising outcomes** (rational individuals, bad collective results). *Treat the model as a **map**: it leaves things out precisely so it can be useful.*

Key terms

Term	Meaning
Rational choice	outcomes emerge from actors pursuing goals under constraints
Preferences	an actor's **ranking of outcomes** (x > y = prefers x to y)
Complete	the actor can compare any two outcomes
Transitive	if x > y and y > z then x > z (consistent ranking)
Constraint	what limits feasible options: resources · institutions · information
Incentive	anything that changes the costs/benefits of an outcome
Expected value	probability × payoff (how to compare choices under risk)
Strategic situation	your best choice depends on what others do
Payoff matrix	table of outcomes & payoffs for a 2-player simultaneous game
Best response	your best strategy given what the others do
Nash equilibrium	a profile where everyone best-responds → no one deviates
Prisoner's dilemma	individually rational choices → collectively worse (non-Pareto) outcome
Bounded rationality	limited info/time → shortcuts & "good enough" (Simon)
Loss aversion	losses loom larger than equal gains → status-quo bias (Kahneman & Tversky)

Exam pointers

- **Set up any analysis** with the three building blocks — actors, preferences, constraints — and state your simplifying assumptions.
- **Define rationality** precisely (complete + transitive) and say what it is **not.**
- **Compute an expected value** and use it to show when a firm cheats — and **which lever** (fine size vs inspection rate) flips the choice.
- **Solve a 2×2 game:** underline best responses, find the **Nash equilibrium**, and explain why it's a good prediction.
- **Identify a prisoner's dilemma** and connect it to Lecture 4.1: the equilibrium is **not Pareto efficient.**
- **Name two critiques** (bounded rationality; loss aversion) and the "model as map" defence.

★ Fun facts & memorable details (Tools 4.2: Rational Choice)

> The sticky bits from the rational-choice lecture.

- **"Rational" doesn't mean selfish or smart.** It only means your preferences are **consistent** (complete + transitive). A selfless, ideological, mistaken person can be perfectly "rational."
- **Your vote won't decide the election — 1 in 60 million.** Yet most people vote. The resolution: voting is about **expression and duty**, not just the result — and that's a real preference, not a mistake.
- **"Clean air is good" changes nothing.** A rule only works if actors **choose** to obey it — so you need a fine big enough (or inspections frequent enough) that the **expected** cost beats the gain.
- **A good policy that fails on purpose.** The height-ban reform looked sensible, but a simple 2×2 game shows every city rationally picks "Don't" — the shortage survives. Models earn their keep by catching this.
- **The trade war is a prisoner's dilemma.** US–China 2025: both slapped on tariffs (the Nash equilibrium) even though mutual free trade would make both richer. Individually rational, collectively poorer.
- **Losses hurt twice as much as gains feel good.** Kahneman & Tversky — which is why cutting a benefit enrages people more than never granting it, and why the status quo is so sticky.
- **Treat the model as a map.** A map is useful *because* it leaves things out. Rational choice is wrong in the details and right about the shape — that's the point.
